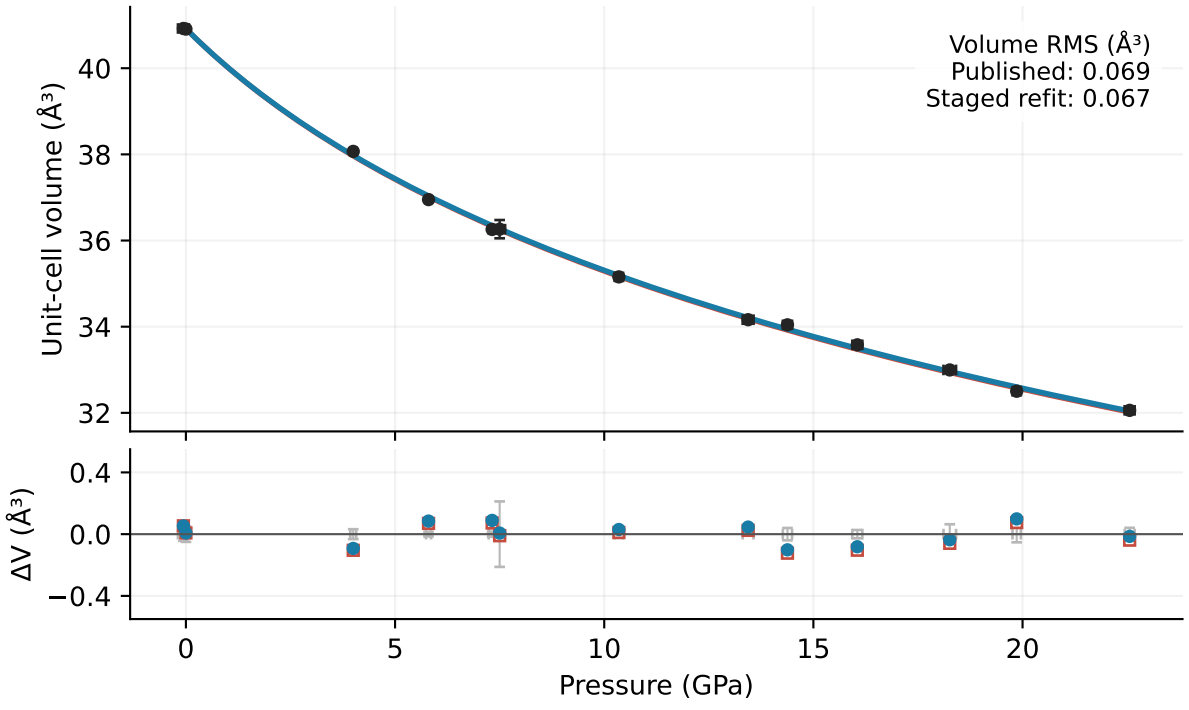
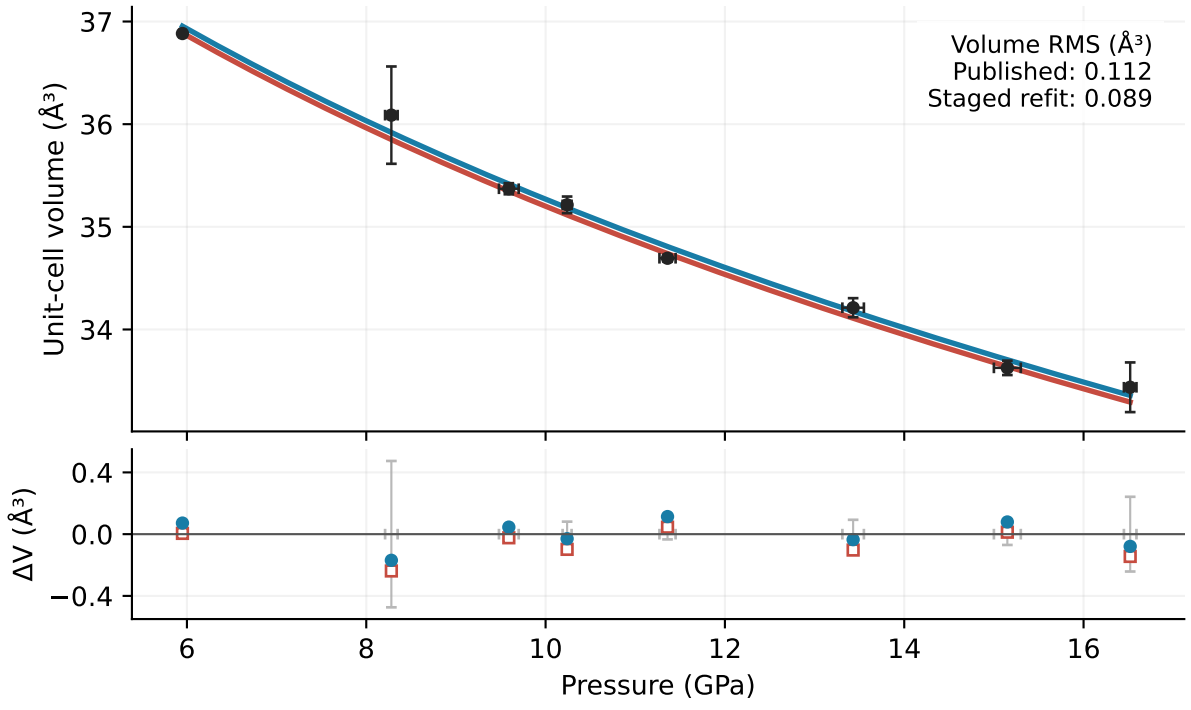
Brucite: staged isotherm refits versus the published EOS

Published Table 4 observations + reported errors
Published expansion law + tabulated isotherm moduli
Staged refit: fitted ambient law fixed; K_0 and K' fitted per isotherm

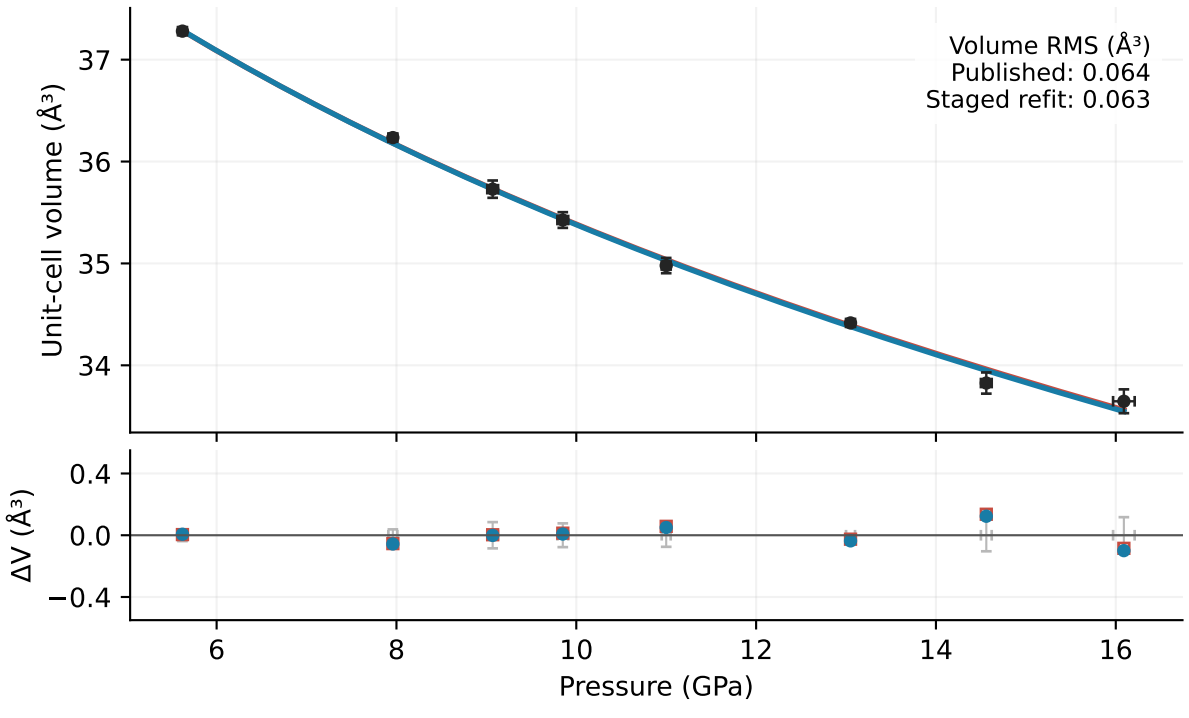
300 K · 13 observations



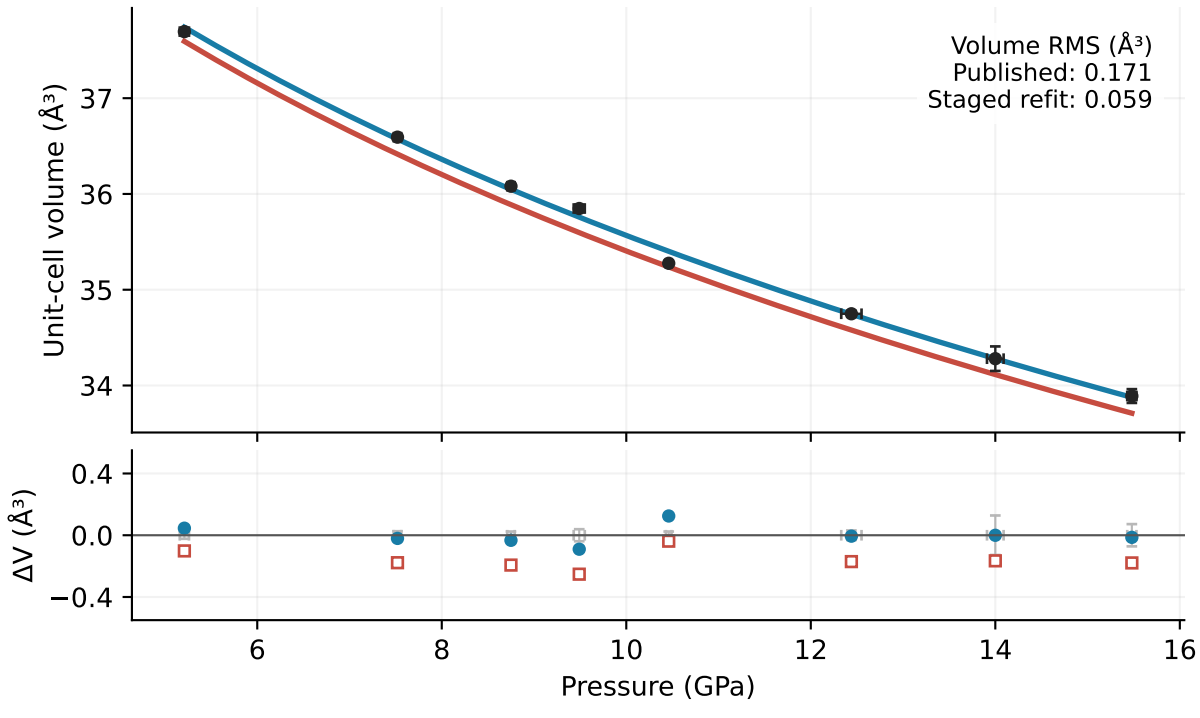
473 K · 8 observations



673 K · 8 observations



873 K · 8 observations



Stage 1: unweighted fit to 11 ambient volumes gives V_{300} = 40.916585 Å³ and constant α = 6.6918518e-05 K⁻¹.
Stage 2: hold $V_0(T)$ fixed; independently fit K_0 and K' using unweighted pressure residuals to all 13/8/8/8 Table 4 rows.
ΔV = calculated − observed volume at observed pressure. Grey bars show reported V/P errors about zero; model uncertainty is not shown.
Four independent isotherms, not a continuous EOS. Original author objective and weights are unreported; published coefficients are preserved.